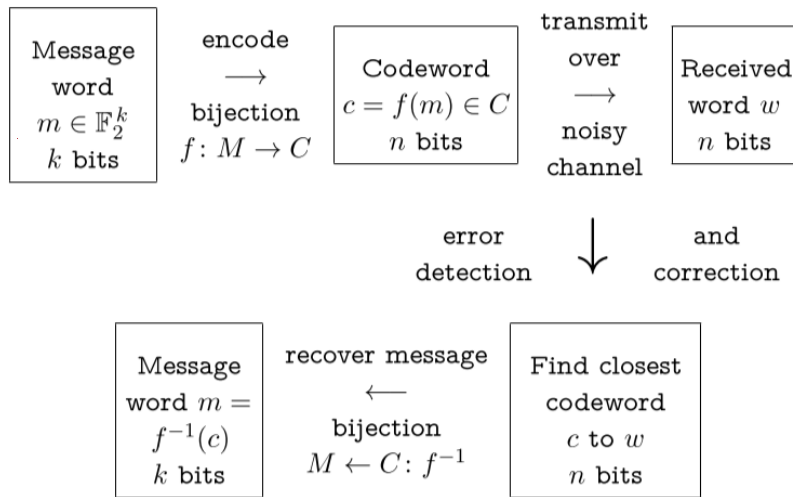


Summary from Monday:



k - length of message words (dimension) $M = \mathbb{F}_2^k$

n - length of codewords $C \subseteq \mathbb{F}_2^n$ $|C| = 2^k$

δ - min. distance between any 2 codewords

An (n, k, δ) -code is $(\delta-1)$ -error detecting.

An (n, k, δ) -code is $\left\lfloor \frac{\delta-1}{2} \right\rfloor$ -error correcting.

An (n, k, δ) -code has rate $\frac{k}{n}$.

An (n, k, δ) -code has relative distance $\frac{\delta}{n}$.

For a binary symmetric channel with reliability p ,
 $P_p(c \rightarrow w) = p^{n-d(c,w)} (1-p)^{d(c,w)}$ is the probability that
 w is received if c is transmitted

Exercise: Suppose that 10^6 bits per second are transmitted over a BSC with reliability $p = 0.999999$. What is the expected frequency of words with undetected errors if:

- (a) The trivial code of all words in $\mathbb{F}_2^8$ is used?
 (b) The parity check code of dimension 8 is used, that is, the code consisting of all even weight words in $\mathbb{F}_2^9$?

(a) Every error is undetected.

$P(\text{undetected error in any word})$

$$= 1 - P(\text{no error in word})$$

$$= 1 - (0.999999)^8$$

$$= 1 - (1 - 10^{-6})^8$$

$$\approx 1 - (1 - 8 \cdot 10^{-6})$$

$$= 8 \times 10^{-6}$$

useful approximation

$$(1+x)^n \approx 1 + nx \text{ for small } |x|.$$

Since

$$(1+x)^n = \binom{n}{0}x^0 + \binom{n}{1}x^1 + \underbrace{\binom{n}{2}x^2 + \dots + \binom{n}{n}x^n}_{\text{small}} \approx 1 + nx$$

Transmission rate:

$$\frac{10^6 \text{ bits}}{\text{second}} \cdot \frac{1 \text{ word}}{8 \text{ bits}} = \frac{10^6}{8} \text{ words/second}$$

$$\# \text{ undetected incorrect words per second} = \frac{10^6}{8} \times 8 \times 10^{-6} = 1$$

(b) probability of undetected error in word

$$\leq 1 - P(0 \text{ errors}) - P(1 \text{ error})$$

$$= 1 - p^9 - 9p^8(1-p)$$

$$= 7.2 \times 10^{-11}$$

Transmission rate

$$\frac{10^6 \text{ bits}}{\text{sec.}} \cdot \frac{1 \text{ word}}{9 \text{ bits}} = \frac{10^6}{9} \text{ words/sec}$$

$$\frac{10^6}{9} \times 7.2 \times 10^{-11} = 8 \times 10^{-6} \text{ undetected incorrect words per second}$$

(≈ 1 undetected incorrect word per 34.72 hrs)

With no p.c.

1 undetected error per second

$$\frac{10^6}{8} \approx 125000 \text{ words per sec.}$$

With p.c.

1 undetected error per 34.72 hrs

$$\frac{10^6}{9} \approx 111111 \text{ words per second}$$

3.5.4 Theorem Suppose communication is via a BSC with reliability p with $0.5 < p < 1$. Let c_1 and c_2 be codewords of length n , and let w be any word of length n . Then

$$\mathbb{P}_p(c_1 \rightarrow w) \geq \mathbb{P}_p(c_2 \rightarrow w) \iff d(c_1, w) \leq d(c_2, w).$$

In other words, the closer a codeword is to w , the more likely it is that it was the word sent.

Proof First note that $0.5 < p < 1 \implies 0 < 1 - p < p$. Hence $\frac{p}{1-p} > 1$. Let $d_1 = d(w, c_1)$, $d_2 = d(w, c_2)$. Then:

$$\begin{aligned} \mathbb{P}_p(c_1 \rightarrow w) &\geq \mathbb{P}_p(c_2 \rightarrow w) \\ \iff p^{n-d_1} (1-p)^{d_1} &\geq p^{n-d_2} (1-p)^{d_2} && \text{by Equation (3.5.2)} \\ \iff \left(\frac{1-p}{p}\right)^{d_1} &\geq \left(\frac{1-p}{p}\right)^{d_2} \\ \iff \left(\frac{1-p}{p}\right)^{d_1-d_2} &\geq 1 \\ \iff \left(\frac{p}{1-p}\right)^{d_2-d_1} &\geq 1 \\ \iff d_2 &\geq d_1 && \text{because } \frac{p}{1-p} > 1. \quad \square \end{aligned}$$

Since $d(c_1, w) = \|c_1 + w\|$, we can restate Theorem 3.5.4 as

$$\mathbb{P}_p(c_1 \rightarrow w) \geq \mathbb{P}_p(c_2 \rightarrow w) \iff \|c_1 + w\| \leq \|c_2 + w\|.$$

The most likely codeword sent is the one with the error pattern of smallest weight.

3.6. Overall Success (Q_C) and Fail (F_C) Probabilities

Let C be a code transmitted over a BSC. Let $c_0 \in C$.

- ★ Let $Q_C(c_0)$ denote the probability that if c_0 was sent, the received word will be successfully corrected back to c_0 .
- ★ Let Q_C be the average³ of all the $Q_C(c_0)$ as c_0 varies across C . This is a measure of the overall success rate of C .

The overall failure probability of the code is

$$F_C = 1 - Q_C.$$

Example (in notes). Determine Q_C for the 3-fold repetition code $C = \{000000, 010101, 101010, 111111\}$.

Start by determining $Q_C(000000)$.

Which received words would be "corrected" to 000000.

- 000000 ✓
- any word of weight 1 ✓
- 9 words of weight 2 except for these 6

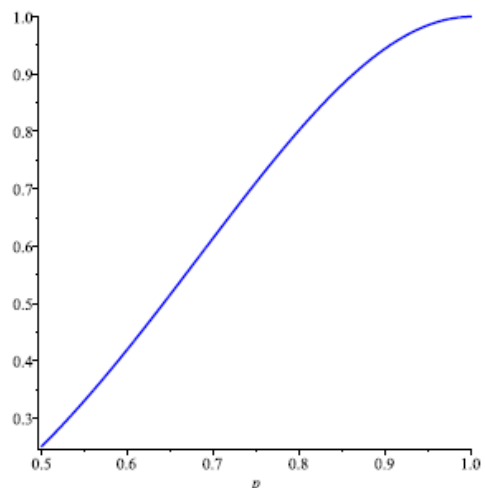
would not be corrected to 000000

→ {
010100
010001
000101
101000
100010
001010

$$Q_C(000000) = p^6 + 6p^5(1-p) + 9p^4(1-p)^2$$

$$= p^4(2p-3)^2$$

Q_C



Exercise Consider the code

$$D = \{0000, 1000, 0111, 1111\}.$$

- (a) Determine Q_D in terms of the channel reliability p .
- (b) What is the minimum reliability p of the channel that would provide an overall failure rate F_D of less than 0.01?
- (a) Cannot correct any weight 2 or more error pattern.
Cannot correct error pattern 1000.
Can correct error pattern 0100, 0010, 0001.

$$Q_D = p^4 + 3p^3(1-p)$$

- (b) Solve $p^4 + 3p^3(1-p) \geq 0.99$
 $p \geq 0.990279$

3.7. Shannon's Theorem

Suppose we have a BSC with reliability $p > 0.5$. We would like to be able to produce codes that can be used to transmit information across the channel such that the failure probability F_C can be made arbitrarily small. More precisely, we would like a family of codes $C_1, C_2, \dots$ of blocklengths $n_1 < n_2 < \dots$ such that the failure probabilities $F_{C_1}, F_{C_2}, \dots \rightarrow 0$ as $n \rightarrow \infty$. This allows us to pick a suitable code to achieve whatever accuracy is required.

Of course, this is possible by taking n huge compared to k , so there is a gigantic amount of redundancy and extremely low rate $r \rightarrow 0$ as $n \rightarrow \infty$. This is not very interesting.

Instead, we would like to know that we can achieve such a family of codes with r staying fixed as $n \rightarrow \infty$. So that, with only a constant percentage overhead, we can achieve failure probability arbitrarily close to 0. It is not obvious that this is possible.

Shannon's Theorem says that it is possible, provided r is not required to be too large. We say that the channel capacity R of the BSC is the largest rate for which it is possible. More precisely, the channel capacity R of the BSC is the supremum of all rates r for which there exists a family of codes $C_1, C_2, \dots$ all of rate $\geq r$, of strictly increasing blocklengths, such that $F_{C_n} \rightarrow 0$ as $n \rightarrow \infty$.

Roughly speaking, we can transmit information at a rate slightly less than R across our BSC, with essentially no errors (we can arrange that the probability F_C of failure is $< \epsilon$). But with rate slightly above R we cannot achieve this goal, no matter how large we make n . So R is the intrinsic limit of what is possible to achieve with this channel.

3.7.1 Theorem [Shannon, 1948] The channel capacity of a BSC with reliability p exists, and is given by

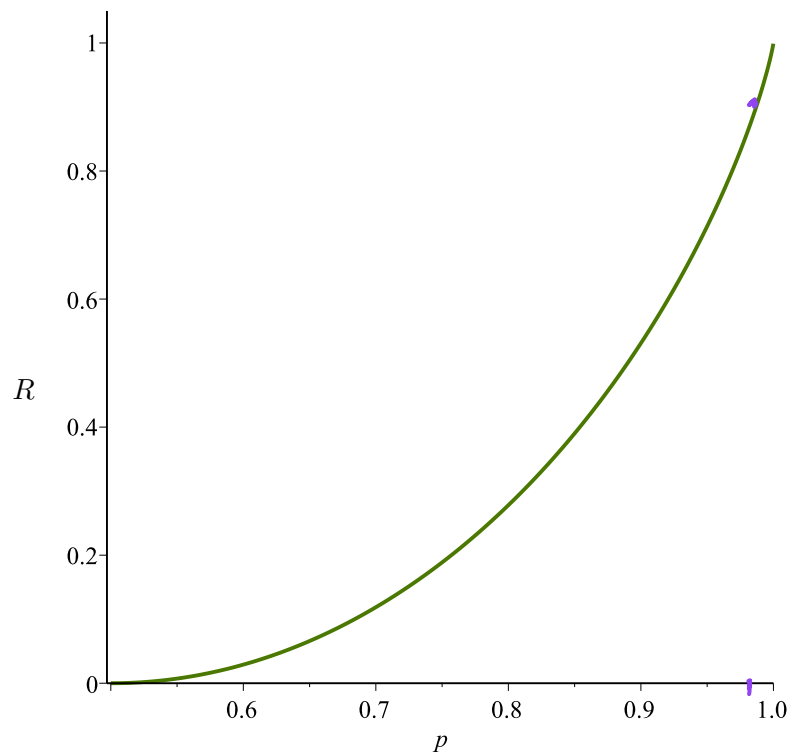
$$R = 1 + p \log_2(p) + (1 - p) \log_2(1 - p).$$

Example Suppose we have a BSC with reliability $p = 0.99$.

Shannon's Theorem says the channel capacity is

$$\begin{aligned} R &= 1 + 0.99 \log_2(0.99) + (0.01) \log_2(0.01) \\ &\approx 0.9192 \end{aligned}$$

Below is a plot of the channel capacity R of a BSC with reliability p , as given by Shannon's Theorem.



Reliability p	0.50	0.66	0.75	0.80	0.85	0.90	0.95	0.98	0.99	0.999
Capacity R	0	0.075	0.189	0.278	0.390	0.531	0.714	0.859	0.919	0.989

Linear Codes I

If we translate all the words in C by a fixed word, then the distance between codewords, and hence the characteristics of C do not change. Thus wlog we may assume $\mathbf{0} \in C$. Now recall Lemma 3.2.8, which says $(C + C) \setminus \{\mathbf{0}\}$ is the set of non-zero error patterns that C cannot detect. We would like to minimize this set.

Since $\mathbf{0} \in C$, $\mathbf{0} + c = c \in C + C$ for every $c \in C$, so $C + C \supseteq C$. So the best (smallest) we could hope for is $C + C = C$, which says C should be *closed under addition*. This means that C will be a *vector space* over $\mathbb{F}_2$, and $f: \mathbb{F}_2^k \rightarrow \mathbb{F}_2^n$ should be an injective map between vector spaces. The most natural choice for f would be a *linear map*. Then f will be described by a matrix of full rank.

4.1.1 Definition A code C is a *linear code* if the sum of any two codewords is also a codeword. That is if $v, w \in C$, then $v + w \in C$. We shall assume $C \neq \{\mathbf{0}\}$, ie that $k > 0$.

4.1. Review of Linear Algebra

We can do linear algebra over $\mathbb{F}_2$ in the same way as over $\mathbb{R}$ or $\mathbb{C}$. All of the same methods and results still apply: row reduction, determinants etc.

Recall that a non-empty subset $U \subseteq \mathbb{F}_2^n$ is a $\mathbb{F}_2$ -vector space iff U is closed under addition and scalar multiplication. Since the only scalars are 0, 1 $\in \mathbb{F}_2$, and $0 \cdot v = \mathbf{0}$, U is a vector space iff $[\mathbf{0} \in U \text{ and } U \text{ is closed under addition}]$. If $U \subseteq \mathbb{F}_2^n$ is non-empty and closed under addition, then U must contain the zero vector (since if $u \in U$, then $u + u = \mathbf{0} \in U$), so will be an $\mathbb{F}_2$ -vector space.

Thus a linear code is exactly a vector subspace of $\mathbb{F}_2^n$.

A *linear combination* of vectors $v_1, \dots, v_n$ is an expression $\sum \alpha_i v_i$, where the α_i are scalars. Over $\mathbb{F}_2$ the only scalars are 0 and 1, so a linear combination of vectors is just a sum of some of them: v_4 , $v_1 + v_3 + v_5$, $v_1 + \dots + v_n$, etc.

A set of vectors $\{v_1, \dots, v_n\}$ is *linearly independent* if no linear combination of them is equal to $\mathbf{0}$, except the trivial combination (empty sum) $0v_1 + \dots + 0v_n$. Since the only scalars are 0 and 1, $S = \{v_1, \dots, v_n\}$ is linearly independent iff no non-empty sum of v_i adds to $\mathbf{0}$. If a set of vectors is not linearly independent, then the set of vectors is said to be *linearly dependent*.

Exercise: Let u, v, w be non-zero vectors in $\mathbb{F}_2^n$.

(a) When is $\{u, v\}$ linearly dependent?

(b) For distinct u, v, w , when is $\{u, v, w\}$ linearly dependent?

(c) Find a set of four linearly dependent vectors such that every proper subset of the vectors is linearly independent.

(a) when $u = v$

(b) when $u + v = w$

(c) $\{100, 010, 001, 111\}$

The span of $S = \{v_1, \dots, v_n\}$, denoted $\text{span}(S)$ or $\langle S \rangle$ is the collection of all linear combinations that can be made from S . This is always a vector space.

A basis for a vector space V is a subset $B \subseteq V$ such that:

★ $\langle B \rangle = V$ (B spans V), and

★ B is linearly independent.

Every vector space V has a basis. In fact, a vector space usually has many different bases, but all bases of V have the same number of elements. The *dimension* of a vector space is the number of elements in (any) basis.

Exercise Let $S = \{1101, 1001, 0101, 0100\}$.

(a) Determine $\langle S \rangle$.

(b) Find a basis for $\langle S \rangle$.

(a) $\langle S \rangle = \{1101, 1001, 0101, 0100, 0000, 0001, 1000, 1100\}$

(b) Dimension 3 $\{1000, 0100, 0001\}$

We also consider matrices with entries in $\mathbb{F}_2$.

Abbreviations:

- ★ ero = elementary row operation.
- ★ ref = row echelon form of a matrix.
- ★ rref = reduced row echelon form (after Gauss-Jordan reduction).

each row has a leading one further to the right than previous rows

leading ones are the only non-zero elements in their column

The only eros are: adding two rows, and swapping two rows.

4.1.3 Theorem Let A be an $m \times n$ matrix with entries in $\mathbb{F}_2$. All of the following quantities are then equal:

- ★ The number of linearly independent rows in A
- ★ The dimension of the space spanned by the rows of A (row space)
- ★ The number of linearly independent columns in A
- ★ The dimension of the space spanned by the columns of A (column space)
- ★ The number of non-zero rows in any ref of A .

Exercise Find the rref of $\begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \end{pmatrix}$ in $\mathbb{F}_2$.

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

span $\{10111, 11010, 01101, 11100\}$ has dimension 3

4.1.4 Definition The number of linearly independent rows of a matrix A is called the rank of A . This is equal to the other quantities listed in Theorem 4.1.3. The set of all v such that $Av = \mathbf{0}$ is called the *nullspace* or *kernel* of A , and denoted $\text{NS}(A)$.

4.1.5 Theorem Let A be an $m \times n$ matrix with entries in $\mathbb{F}_2$. Then

$$\text{rank}(A) + \dim \text{NS}(A) = n.$$

4.2. Basic Properties of Linear Codes

One of the advantages of a linear code is that its distance is easy to find.

4.2.1 Theorem For any linear code $C \neq \{0\}$, the distance of C is equal to the smallest weight of any nonzero codeword.

Proof Let C be a linear code of distance δ , and let c_0 be a nonzero codeword of smallest weight. Since $0, c_0 \in C$ the distance of C is at most $d(c_0, 0) = \|c_0\|$, so $\delta \leq \|c_0\|$. Suppose $\delta < \|c_0\|$. Then there are two codewords $c_1, c_2 \in C$ such that $\delta = d(c_1, c_2) < \|c_0\|$. As C is linear, $c_1 + c_2$ must also be a codeword, of weight $\|c_1 + c_2\| = d(c_1, c_2) < \|c_0\|$. But this contradicts the assumption that c_0 is the nonzero codeword of smallest weight. $\square$

It is easy to find the distance of a linear code. Other advantages of linear codes include:

- ★ For linear codes, there is a procedure for IMLD which is simpler and faster than we have seen so far (for some linear codes, there are very simple decoding algorithms).
- ★ Encoding using a linear code is faster than for arbitrary non-linear codes.
- ★ It is easy to manufacture linear codes. Just take the span of any set of vectors.
- ★ It is easy to describe the set of error patterns that a linear code will detect.
- ★ It is much easier to describe the set of error patterns a linear code will correct than it is for arbitrary non-linear codes.

Disadvantage: linear codes are not necessarily optimal in terms of their error correcting ability.

Recall that an (n, k, δ) code has 2^k codewords of length n , and minimum distance δ .

A *basis* for a linear code C is a basis for the vector space C . The *dimension* of C is its dimension as a subspace of $\mathbb{F}_2^n$, namely the number of elements in a basis.

For a linear code we have now given two different definitions of dimension. However they agree: Suppose a linear code C has dimension k in the vector space sense. Let $B = \{c_1, c_2, \dots, c_k\}$ be a basis for C . Then each codeword b in C can be written as

$$b = a_1c_1 + a_2c_2 + \dots + a_kc_k$$

for a unique choice of scalars $a_1, a_2, \dots, a_k$. Since each a_i is 0 or 1 for $1 \leq i \leq k$, there are 2^k distinct choices for $a_1, a_2, \dots, a_k$, and hence for b .

So the dimension of C in the linear algebra sense agrees with our definition of the dimension k of a code in the general case.

4.3. Bases for Codes

How can we describe a vector space C ? Two approaches are: give a basis, or give a set of equations defining the elements of C .

If $c_1, \dots, c_k$ forms a basis of C , then every element $b \in C$ can be written uniquely in the form $b = a_1c_1 + \dots + a_kc_k$. Let $\mathbf{G}$ be the matrix formed by taking row i to be c_i for $1 \leq i \leq k$. Then expressions $a_1c_1 + \dots + a_kc_k$ are exactly the vectors obtained by multiplying $\mathbf{G}$ by the vector $(a_1, a_2, \dots, a_k)$. That is, C is the image of the matrix $\mathbf{G}$.

Alternatively, one can describe C as all vectors satisfying some linear equations, such as $C = \{x \mid x_1 + x_3 = 0, x_2 + x_4 = 0\}$ etc, where x is the word with bits $x_1x_2\dots$. This says x is an element of the nullspace of some matrix $\mathbf{H}$.

That is, C may be defined as the image of one map ($\mathbf{G}$) or the kernel of another ($\mathbf{H}$).

In detail, we shall show that associated to each linear code C are two matrices, denoted $\mathbf{G}$ and $\mathbf{H}$:

- ★ $\mathbf{G}$ is used for encoding.
- ★ $\mathbf{H}$ is used for error correction/detection and for finding the distance δ .

Eg. code $\{0000, 1000, 0111, 1111\}$
 Dimension 2
 Basis: $\{1000, 0111\} \Rightarrow$ create a generating matrix $\mathbf{G}$
 Equationally defined: $x_2 + x_3 = 0$
 $x_2 + x_4 = 0$
 create a parity-check matrix $\mathbf{H}$ $x_2 = x_3 = x_4$
 code words x_1, x_2, x_3, x_4

```

      000      111
       / \    / \
    0000 1000 0111 1111
    
```

First, recall the method of finding a basis for a linear code (vector space):

4.3.1 Algorithm [Algorithm for finding a basis for $C = \langle S \rangle$.]

Let S be a non-empty subset of $\mathbb{F}_2^n$.

- ★ Form the matrix A whose rows are the words in S .
- ★ Use eros to find a ref (or rref) of A .
- ★ The non-zero rows of the ref of A form a basis for $C = \langle S \rangle$.

Exercise : Find a basis for the linear code $C = \langle S \rangle$ where $S = \{0110, 1100, 1001, 0101\}$.

$$\begin{pmatrix} 1100 \\ 0110 \\ 1001 \\ 0101 \end{pmatrix} \rightarrow \begin{pmatrix} 1100 \\ 0110 \\ 0101 \\ 0101 \end{pmatrix} \rightarrow \begin{pmatrix} 1100 \\ 0110 \\ 0011 \\ 0000 \end{pmatrix} \rightarrow \begin{pmatrix} 1001 \\ 0101 \\ 0011 \\ 0000 \end{pmatrix}$$

ref rref

Basis $\{1100, 0110, 0011\}$
or $\{1001, 0101, 0011\}$

code C

$$n = 4$$

$$k = 3$$

$$d = 2$$

$$|C| = 2^3$$

generating matrix $G = \begin{pmatrix} 1100 \\ 0110 \\ 0011 \end{pmatrix}$

4.4. Generating Matrices

We now use the material from the previous section to create a matrix that is used in the encoding process for a linear code.

4.4.1 Definition If C is a linear code of length n and dimension k , then any matrix whose rows form a basis for C is called a generating matrix for C . Such matrices must have k rows and n columns, and have rank k .

$(\quad)_{k \times n}$

Since a vector space may have many different bases, the matrix G is not unique. (It is unique if we insist that it is in rref.)

4.4.2 Theorem A $k \times n$ matrix G with $n \geq k$ is a generating matrix for some (n, k) linear code C iff the rows of G are linearly independent (so the rank of $G = k$).

4.4.3 Theorem If G is a generating matrix for a linear code C , then any matrix row equivalent to G is also a generating matrix for C . In particular, every linear code has a unique generating matrix in rref.

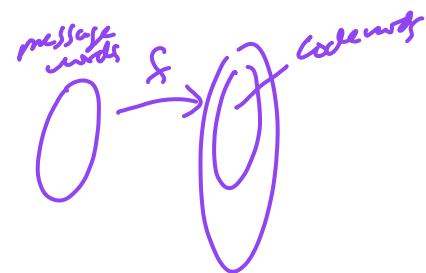
4.4.5 Example Let C be the linear code $C = \{000000, 001110, 010101, 011011, 100011, 101101, 110110, 111000\}$. Show that G below is a generating matrix for C .

$n = 6$
 $k = 3$

$$G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

Solution: Use Algorithm 4.3.1. Or note that

- ★ $|C| = 8$ so $\dim C = 3$, and G is 3×6 ,
- ★ The rows of G are linearly independent,
- ★ Each individual row of G is a codeword.



So the rows of G generate a 3-dimensional subspace of the 6-dimensional space C , and hence must exactly generate C .

matrix multiplication
 $1 \times k \quad k \times n$

4.4.6 Theorem If G is a generating matrix for a linear (n, k) -code C , then $c = mG$ ranges over all 2^k codewords in C as m ranges over all 2^k words of length k . Thus $C = \{mG \mid m \in \mathbb{F}_2^k\}$. Moreover, $m_1G = m_2G$ iff $m_1 = m_2$.

Message m is encoded as $c = mG$, where m and c are represented by row vectors.

Example: Using $G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix}$ for code C from Example 9.4.5,

encode the message word 011.

$$(011) \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix} = (011011)$$

← easier to recover the message word from the codeword.

$G' = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{pmatrix}$ is another generating matrix for C .

Encode the message word 011 using G' .

$$(011) G' = (001110)$$

$$\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \uparrow & \uparrow & & \uparrow \end{array}$$

4.5. Systematic and Equivalent Codes

4.5.1 Definition Let $\mathbf{G}$ be a $k \times n$ matrix with $k \leq n$. We say $\mathbf{G}$ is in *standard form* if the first k columns of $\mathbf{G}$ form the $k \times k$ identity matrix $\mathbf{I}_k$. That is, $\mathbf{G} = (\mathbf{I}_k \mid \mathbf{X})$. Any such $\mathbf{G}$ automatically has linearly independent rows and is in rref. Thus $\mathbf{G}$ is a generating matrix for some linear code of length n and dimension k . We say that the code C generated by $\mathbf{G}$ is *systematic*.

Not all linear codes have a generating matrix in standard form.

4.5.2 Example Is the linear code $C = \{00000, 10110, 10101, 00011\}$ systematic?

Solution: We find a generating matrix. There is no need to write down the $\mathbf{0}$ word. ($\mathbf{0}$ is never part of a basis.) So:

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix} \rightarrow$$

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ rref}$$

$C = \{00000, 11100, 10101, 01001\}$

So $\mathbf{G} = \left(\begin{array}{cc|cc} 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{array} \right)$. The unique rref is not in standard form, so C is not systematic. $\otimes$



swap bits 2 and 4 in each codeword

4.5.3 Theorem If C is a linear code of length n and dimension k with generating matrix $\mathbf{G}$ in standard form and a message m is to be transmitted, then the first k bits of the codeword $v = m\mathbf{G}$ form the message word $m \in \mathbb{F}_2^k$.

4.5.5 Definition If C is any code of length n , we can obtain a new code C' of length n by applying a fixed permutation σ of the n bits. The resulting code $C' = \sigma(C)$ is said to be *equivalent* to C .

4.5.7 Theorem Any linear code C is equivalent to a linear code having a generating matrix in standard form.

Exercise Consider the $(6,3)$ linear code with generating matrix $G = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$

- (a) Determine the codeword corresponding to the message word 011.
- (b) Determine the message word corresponding to the codeword 101101.
- (c) What is the distance of this code?

(a) 011110

(b) 101

(c) $d = 3$